

EXERCISE 2

At first, I started creating the Karnaugh map for 4 variables, indicated with the numbers in the corners. Later, I wrote every minterm in its specific case (0, 1, 2, 3, 4, 7, 11, 15). Finally, I formed two groups of 4: one ($a'b'$) and another (cd), as well as one group of 2 ($a'c'd'$).

$ab \backslash cd$	00	01	11	10
00	1	1	1	1
01	1		1	
11			1	
10			1	

$S = \bar{a}\bar{b} + cd + \bar{a}c\bar{d}$